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Subject: PL Roll No.: 18 Exp. No.: _____

Name of the Experiment: _____

Marks

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with date

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Performed on: _____

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Q.1 List the python sequence data types and state one key characteristic of each with respect to mutability or ordering.
In python, sequence data types store collection of items into a specific order.

1. List : • Ordered sequence
• Mutable (elements can be changed after creation)

2. Tuple : • Ordered sequence
• Immutable (cannot be changed once created)

3. String : • Ordered sequence of characters
• Immutable (characters cannot be modified)

4. Range : ordered sequence of no.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐

Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

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2 Explain how indexing and slicing work in python sequences. Illustrate how they differ when applied to string and list.

Indexing in python sequence
Indexing is used to access a single element from a sequence

Indexing starts from 0 (left to right)

Negative indexing starts from -1 (right to left)

Ex :

Code

```
s = "python"
```

```
l = [10, 20, 30, 40]
```

```
print (s[0])
```

```
print (s[-1])
```

```
print (l[1])
```

```
print (l[-2])
```

Slicing in python sequence

Slicing is used to extract a part (subsequence) of a sequence.

Syntax :-

Sequence [start : stop : step]

Start → starting index (included)

Stop → ending index (excluded)

Step → gap between elements (optional)

```
print (S [1:4])  
print (S [0:3])  
print (S [::2])  
print (S [::-1])
```

Q.3 Write a python program that
• Uses indexing and slicing on a string
or list, and

Apply at least two built-in function
to display meaningful result.

Python code

```
text = "python programming"
```

indexing

```
first_char = text [0]
```

```
last_char = text [-1]
```

slicing

```
sub_text = text [0:6]
```

```
every_second = text [::2]
```

```
length = len (text)
```

```
max_char = max (text)
```

```
print ("original string:", text)
```

```
print ("first character:", first_char)
```

```
print ("last character:", last_char)
```

```
print ("sliced string (0:6):" sub_text)
```

```
print ("Every second char : " every-second)
print ("length of string : " length)
print ("Maximum char : ", max-char)
```

4 Analyze the difference betⁿ equality (==) and identity list used with sequence types. Discuss how objects mutability and memory reference affect program behaviour.

Difference between == and is in python. The equality operator (==) compares the memory location of two objects. The identity operator (is) compares the memory location of two objects.

Two sequence may have the same value but be stored in different memory location.

So == returns true while the return is false.

For mutable sequence like list, if two variable refer to the same objects changes made through one variable affect the other.

For immutable sequence like string and tuple, values cannot be modified any changes creates a new object reducing side effects.

5 Evaluate the suitability of lists, sets, sets and dictionaries for storing and managing unique student records. Justify.

Suitability of list, set and dictionary for unique student records.

When storing and managing unique student records structure affects efficiency, data integrity and ease of access.

1. List:

List store elements in an ordered sequence. They allow duplicate entries. So extra logic is required to ensure uniqueness. Searching for a student record is slow ($O(n)$) specially for larger datasets.

2. Sets:

Sets store only unique elements automatically they are unordered and do not support indexing. Set membership checking ($O(1)$) avg (time).

Dictionary store the data as Key value pairs. Keys are unique, ensuring no duplicate student IDs.

Very fast access, insertion and deletion ($O(1)$) average time.

Can store structured record as values.

Ques. 6 Design and implement a python program that uses

```
Student = {  
1 : {"name": "Amit", "Marks": [75, 80, 95]},  
2 : {"name": "Sneha", "Marks": [88, 91, 81]}  
}
```

You roll data in student.items():
marks = data["Marks"]

first_marks = marks[0]

top_two = sorted(marks, reverse=True)[:2]

total = sum(marks)

average = total / len(marks)

print(f"\n Roll No : {Roll}")

Print(f"name : {data['name']}")

Print(f"marks : {marks}")

Print(f"first Marks : {first Marks}")

Print(f"Top two marks : {top_two}")

Print(f"Total : {total} average : {average}")

Q.7 Write a program to print the first and last character of a string.

text = input("Enter a string: ")

print("First character:", text[0])

print("last characters:", text[-1])

text[0] → access the first character

text[-1] → access the last character using negative indexing.

8. Write a program to create a list tuple, set and dictionary.

```
My_list = [10, 20, 30, 40]
```

```
My_tuple = (1, 2, 3, 4)
```

```
My_set = {5, 10, 15}
```

```
My_dict = { "Name": "Amrit",  
            "age": 20,  
            "branch": "CSF"  
}
```

```
print("list": My_list)  
print("Tuple": My_tuple)  
print("Set": My_set)  
print("Dictionary": My_dict)
```

9. Write a program to understand the concept of indexing.

```
text = "python"  
print("string": text)  
print("First char": text[0])  
print("Second char": text[1])  
print("last char": text[-1])
```

```
numbers = [10, 20, 30, 40]
```

```
print ("In list: ", numbers)
```

10 Write a program to understand the concept of built in function.

```
Numbers = [10, 20, 30, 40, 50]
```

```
print ("list: ", numbers)
```

```
print ("length of list: ", len(Numbers))
```

```
print ("Max value: ", Max(Numbers))
```

```
print ("Min value: ", Min(Numbers))
```

```
print ("Sum of elements: ", Sum(Numbers))
```

```
print ("Sorted list: ", Sorted(Numbers))
```